

Dhruv Mavani

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Summary

Systems and Full-Stack Developer focused on high-performance distributed networks and decentralized architecture. Advanced proficiency in the **Solana web3 ecosystem**, state validation, and writing secure, production-grade smart contracts in Rust. Proven track record of managing complex state logic across concurrent environments and translating distributed workflows into responsive, high-throughput software.

Education

Parul University | CGPA: 7.39/10.0

August 2022 – June 2026

- Bachelor of Technology in **Computer Science and Engineering**.

Technical Skills

Languages: Rust (Anchor Framework), TypeScript, JavaScript

Web and Frameworks: @solana/web3.js, React.js, Next.js, Node.js, HTML, Tailwind CSS

Databases: PostgreSQL (supabase), MySQL, Firestore

Platforms and Tools: Anchor CLI, Git, GitHub, Chrome DevTools, Vercel

Projects

Candl - An On-Chain AMM for NFTs [Live Site](#) [GitHub](#)

July 2026 – Present

- **Tools:** Rust (Anchor), Next.js, TypeScript, PostgreSQL, Vercel
- Built **Candl**, a protocol giving every NFT its own on-chain **bonding-curve AMM** for instant buy/sell liquidity and live price discovery in place of static marketplace listings, implemented across **6+ Rust/Anchor instructions**.
- Diagnosed and fixed a **vault rent-exemption deadlock** blocking small first buys, validated via a **16-test LiteSVM** in-process Rust suite.
- Built a Next.js/TypeScript frontend with real-time **lightweight-charts candlesticks**, SWR-pollled market data, and Solana Wallet Adapter integration; stress-tested with an automated bot-trading harness generating **324 live trades** in 15 minutes.

RustX - A Centralized Exchange [GitHub](#)

Jun 2026 – Jul 2026

- **Tools:** Rust, Actix-web, React.js, Tailwind CSS, Vite, Framer Motion
- Built a high-performance centralized exchange featuring a **Level-3 FIFO Price-Time Priority matching engine** capable of processing full fills, partial fills, and resting orders with sub-millisecond in-memory execution.
- Engineered **6+ REST APIs**, an orderbook with **O(1)** order cancellation, custom SVG market depth visualization, live trade history, and real-time wallet synchronization for buy/sell workflows.
- Developed **7+ frontend modules** including order placement, depth chart, orderbook, trade history, wallet management, and open orders dashboard using React with a Rust backend architecture.

Solana SPL Token Launchpad [Live Site](#) [GitHub](#)

April 2026 – May 2026

- **Tools:** React.js, Solana Wallet Adapter, SPL Token Program, Metaplex, Tailwind CSS, Vercel
- Built a **production-ready Solana launchpad** enabling users to create and deploy SPL tokens on **mainnet and devnet** with wallet integration in under a minute.
- Engineered **5+ deployment modules** including token creation, Metaplex metadata integration, dark mode UI, and multi-network support.

Open Source and Publications

- **anza-xyz/agave: Enforce shred_version matching during deserialization** -> [View Pull Request](#)
 - Merged into Solana's core validator client, fixing a **critical consensus bug** in the `votor-messages` crate.
 - Enforced `shred_version` validation during byte deserialization via a custom `SchemaReadContext`, making the network **safer by construction** and preventing skipped checks downstream.
- **orca-so/whirlpools: Fix JSDoc for calculateMinimumBalanceForRentExemption** -> [View Pull Request](#)
 - Corrected rent exemption documentation in the Whirlpools SDK, improving **developer experience** for integrators building on Orca's **concentrated liquidity AMM**.
- **Research Publication (IJIRT 2026): Bridging Learning, Gamification, and Digital Work** - A review on integrating gamified learning with freelance ecosystems -> [View Paper](#)